ORGANIC CHEMISTRY FRONTIERS

RESEARCH ARTICLE

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Org. Chem. Front.*, 2020, **7**, 1016

# Construction of spirooxindole-fused spiropyrazolones containing contiguous three stereogenic centres *via* [3 + 2] annulation utilizing a ferrocene derived bifunctional phosphine catalyst†

Wenjun Luo, Bingxuan Shao, Jingyi Li, Xiao Xiao, Dingguo Song, Fei Ling and Weihui Zhong *

Received 5th February 2020,
Accepted 18th March 2020

DOI: 10.1039/d0qo00140f

rsc.li/frontiers-organic

A ferrocene-derived bifunctional phosphine-catalyzed enantioselective [3 + 2] annulation of γ-substituted allenoates with pyrazoloneyldiene oxindoles is investigated for the construction of spiro-cyclopentene-linked oxindole and pyrazolone compounds, which bear three consecutive stereocenters and two vicinal spiroquaternary chiral centers, in good chemical yields (up to 95%) with excellent regio-, diastereo- (dr > 19 : 1) and enantioselectivities (up to 98% ee).

Published on 19 March 2020. Downloaded by Fudan University on 2/20/2021 4:53:07 AM.

Spirocyclic compounds, bearing at least one quaternary carbon, are valuable structural mofits due to their broad applications as biomedicinal natural molecules and pharmaceuticals.[1] Among multifarious spirocyclic compounds, spiro-cyclopentane-oxindole (SCPO)[2] and spiro-cyclopentane-pyrazolone (SCPP)[3] have attracted extensive interest from pharmaceutical and organo-synthetic communities due to their myriad biological activities (Fig. 1). For example, SCPOs **I** and **II** have been reported as natural compounds,[4] and **III** is an inhibitor of p53/MDM2.[5] Moreover, SCPPs **IV** and **V** have been recognized as inhibitors of type-4 phosphodiesterase,[6] and **VI** serves as a PPAR antagonist.[7]

The combination of pharmacophores is an effective strategy for discovering new bioactive compounds in pharmaceutical research.[8] Therefore, we are interested in exploring organocatalytic methods to construct spiro-cyclopentene scaffolds linking oxindoles to pyrazolones, which can be potential and trendy tactics for drug discovery. Additionally, it has been envisioned that the phosphine-mediated [3 + 2] annulation reaction between allene as a $C_3$ synthon and an appropriate $C_2$ synthon, activated pyrazoloneyldiene oxindole, is applicable to construct spiro-oxindole cyclopentene-pyrazolone (Scheme 1), since the first phosphine-catalyzed [3 + 2] cycloadditions of allenoates with electron-deficient olens to generate functionalized cyclopentene rings were reported by Lu in 1995.[9]

Mono-, di-, or tri-substituted activated alkenes are routinely utilized as reaction partners with allenes for the synthesis of cyclopentenes,[10] but the utilization of tetra-substituted alkenes was extremely limited. To the best of our knowledge, there was only one limited literature study by Lu, who reported the [3 + 2] annulation of allenes and tetra-substituted alkenes to produce dispirocyclic oxindole structural motifs.[11] Very recently, structurally novel bispiro-pyrazolone-3,3′-oxindoles were synthesized *via* [3 + 2] annulation of Morita–Baylis–Hillman carbonates with pyrazoloneyldiene oxindoles in excellent yields, with diastereospecificity and high enantioselectivity catalyzed by (*S*)-SITCP.[11b] Inspired by the above

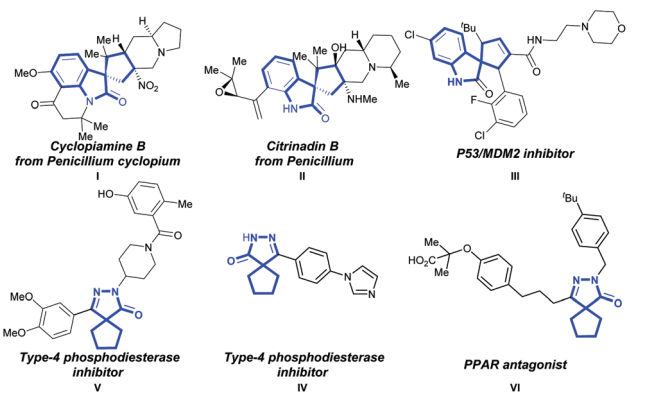



**Fig. 1** The significance of SCPO and SCPP scaffolds.

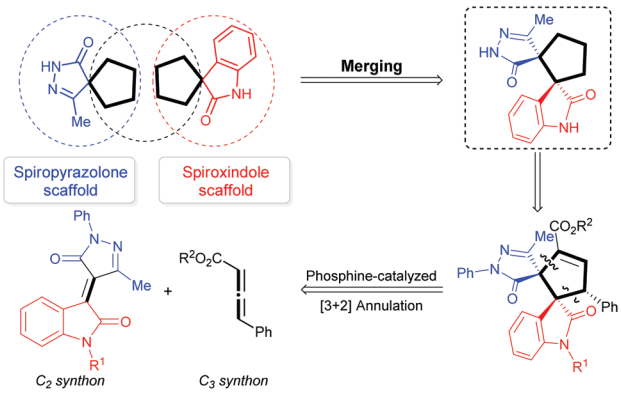
*Key Laboratory for Green Pharmaceutical Technologies and Related Equipment of Ministry of Education, College of Pharmaceutical Sciences, Zhejiang University of Technology, Hangzhou 310014, People's Republic of China.*
*E-mail: weihuizhong@zjut.edu.cn*

†Electronic supplementary information (ESI) available. CCDC 1961000. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d0qo00140f



This journal is © the Partner Organisations 2020

Published on 19 March 2020. Downloaded by Fudan University on 2/20/2021 4:53:07 AM.



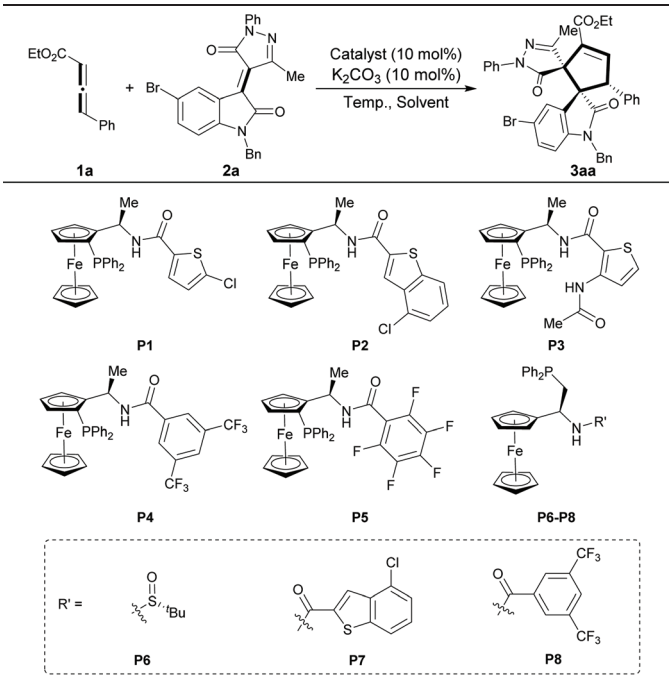

**Scheme 1** Construction of spiro-cyclopentene-linked oxindole and pyrazolone through phosphine-triggered [3 + 2] annulation.

excellent study and our continuing interest in ferrocenyl-phosphine catalysis,[12] we speculated that the spiro-cyclopentene linking oxindole and pyrazolone could be obtained by the [3 + 2] annulation of tetra-substituted activated alkenes derived from oxindole and pyrazolone with γ-substituted allenoates.

To test the feasibility, the [3 + 2] annulation reaction was examined with ethyl 4-phenylbuta-2,3-dienoate **1a** and pyrazoloneyldiene oxindole **2a** in the presence of various phosphine catalysts. According to our previous works,[13] when potassium carbonate was chosen as the additive, the desired [3 + 2] product **3aa** was formed in 90% yield, with high diastereoselectivity (dr > 19 : 1) and regioselectivity (only the γ-adduct) under the catalysis of triphenylphosphine in toluene at room temperature (Table 1, entry 1). Then our previously designed bifunctional chiral ferrocenylphosphines (**P1** and **P2**) were subjected to the [3 + 2] annulation reaction and the desired product **3aa** was obtained with moderate ee values (65% and 68%) and good yields (Table 1, entries 2 and 3), respectively. Next, to verify the efficiency of phosphine catalysts, various ferrocene derived bifunctional phosphine catalysts were investigated (Table 1, entries 4–9). Catalyst **P4** performed well and led to the desired product **3aa** with 75% yield and 81% ee (Table 1, entry 4), which indicated that the 3,5-bis(trifluoromethyl) group on the benzene ring was beneficial for enantioselectivity. Based on the above-mentioned analysis, a further trial of phosphine-ferrocene chiral catalysts was conducted. These results demonstrated that catalyst **P8**, reported by Zhang's group,[14] exhibited more superior reactivity and stereocontrollability, affording the product **3aa** with 85% yield and 87% ee (Table 1, entries 7–9). Gratifyingly, the enantioselectivity was improved to 98% ee, when decreasing the reaction temperature from room temperature to −20 °C (Table 1, entries 10–12). Solvent screenings showed that other solvents failed to give better results than toluene (Table 1, entries 13–15). Thus, the optimal reaction conditions are 10 mol% of **P8** as the catalyst, 10 mol% of potassium carbonate as the additive and toluene as the solvent at −20 °C (Scheme 3).

With the optimal conditions in hand, the substrate scope of both γ-substituted allenoates **1** and pyrazoloneyldiene oxindole components **2** was evaluated. Variation of the $R^1$ group showed that benzene rings with electron-withdrawing or electron-donating groups at the 5-position (Scheme 2, **3aa–3ae**) and the 7-position (Scheme 2, **3af–3am**) gave the desired products with good yields (79–93%) and excellent stereogenic control (90–98% ee). Besides, when $R^1$ was 6-Cl or H, the desired products were also obtained with high yields and ee values (Scheme 2, **3aj** and **3ak**). Subsequently, γ-substituted allenoates **1** with different ester groups on the side chain were studied smoothly. However, the substrate derived from benzyl acrylate afforded a lower ee value than that from methyl acrylate or ethyl acrylate (Scheme 2, **3ak–3ck**).

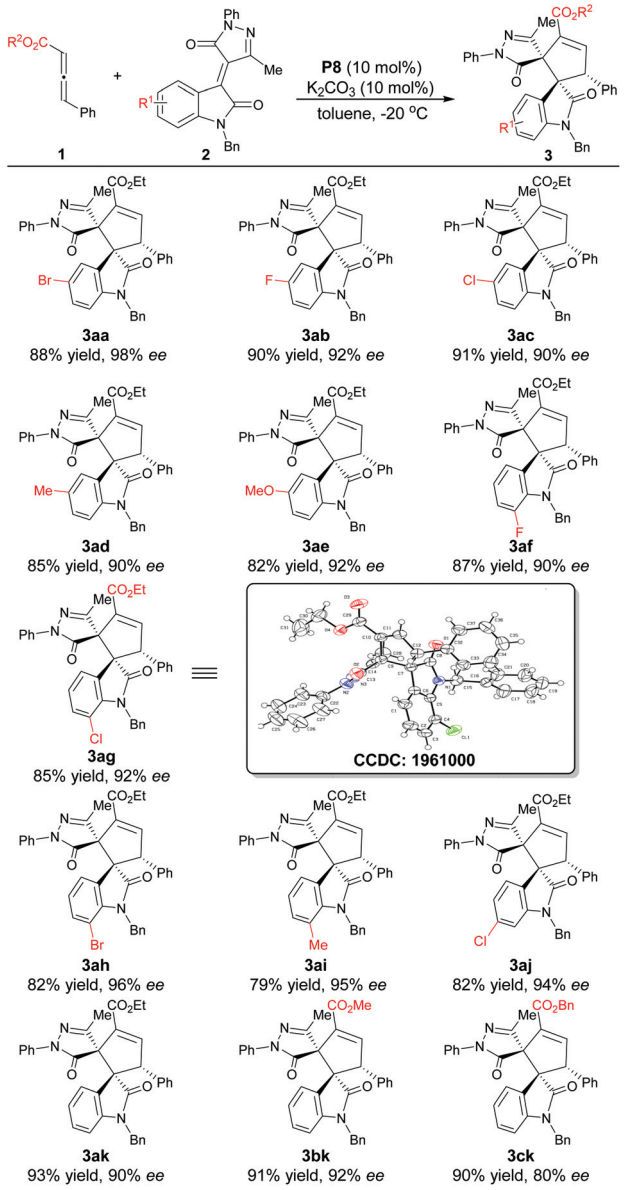
**Table 1** Optimization of reaction conditions[a]




| Entry | Catalyst | Solvent | $T$ (°C) | Yield[b] (%) | ee[c] (%) |
|---|---|---|---|---|---|
| 1 | $PPh_3$ | Toluene | r.t. | 90 | — |
| 2 | **P1** | Toluene | r.t. | 76 | −65 |
| 3 | **P2** | Toluene | r.t. | 78 | −68 |
| 4 | **P3** | Toluene | r.t. | 77 | −41 |
| 5 | **P4** | Toluene | r.t. | 75 | −81 |
| 6 | **P5** | Toluene | r.t. | 76 | 62 |
| 7 | **P6** | Toluene | r.t. | 28 | −60 |
| 8 | **P7** | Toluene | r.t. | 83 | 76 |
| 9 | **P8** | Toluene | r.t. | 85 | 87 |
| 10 | **P8** | Toluene | 0 °C | 85 | 92 |
| 11 | **P8** | Toluene | −10 °C | 86 | 97 |
| **12** | **P8** | **Toluene** | **−20 °C** | **88** | **98** |
| 13 | **P8** | $Et_2O$ | −20 °C | 83 | 95 |
| 14 | **P8** | DCM | −20 °C | 76 | 92 |
| 15 | **P8** | DMF | −20 °C | 20 | 75 |

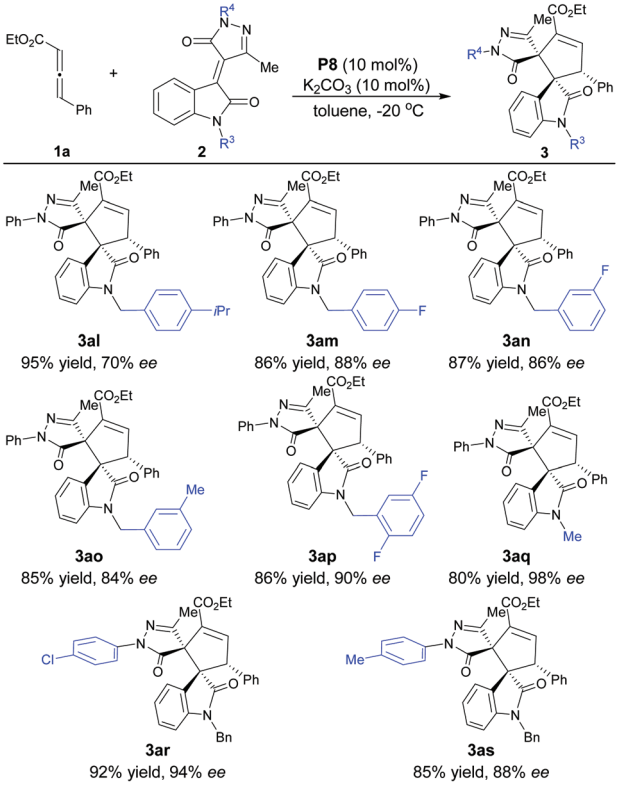
[a] Reaction conditions: **1a** (0.18 mmol), **2a** (0.15 mmol), catalyst (10 mol%), $K_2CO_3$ (10 mol%) in solvent (3.0 mL), 4–6 h, under a $N_2$ atmosphere. [b] Isolated yield, dr value > 19 : 1; the dr values of the products were determined by $^1$H-NMR analysis of the crude product. [c] The ee value was determined by HPLC.

Published on 19 March 2020. Downloaded by Fudan University on 2/20/2021 4:53:07 AM.

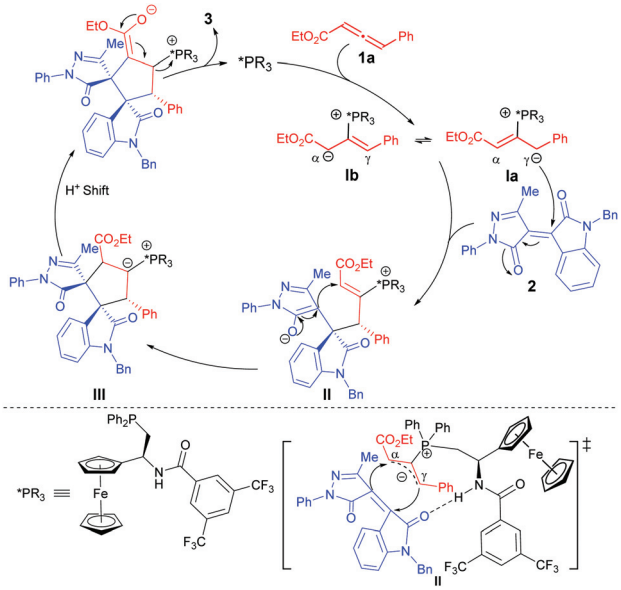



**Scheme 2** Substrate scope of $R^1$ and $R^2$. Reaction conditions: **1** (0.18 mmol), **2** (0.15 mmol), **P8** (10 mol%), $K_2CO_3$ (10 mol%) in toluene, under a $N_2$ atmosphere. Isolated yield, dr > 19 : 1, the dr values of the products were determined by $^1$H-NMR analysis of the crude products. The ee value was determined by HPLC.

Further investigations of both *N*-substituent of the oxindole moiety $R^3$ and *N*-substituent of the pyrazolone moiety $R^4$ were performed with 4-phenyl-2,3-butadienoate **1a**. However, the substituted benzyl group ($R^3$) on oxindole led to lower stereoselectivities than the benzyl group, especially when $R^3$ was 4-isopropylbenzyl (70% ee) (Scheme 2, **3al–3ap**). To our delight, methyl ($R^3$) was well accommodated (Scheme 2, **3aq**). Both the electron-withdrawing and electron-donating groups ($R^4$) at the 1-position on aromatic pyrazolone could provide favourable ee values (Scheme 2, **3ar** and **3as**). In addition, the absolute configuration of **3ag** was confirmed by X-ray crystallography (CCDC 1961000†).[15]




**Scheme 3** Substrate scope of $R^3$ and $R^4$. Reaction conditions: **1a** (0.18 mmol), **2** (0.15 mmol), **P8** (10 mol%), $K_2CO_3$ (10 mol%) in toluene, under a $N_2$ atmosphere. Isolated yield, dr > 19 : 1, the dr values of the products were determined by $^1$H-NMR analysis of the crude products. The ee value was determined by HPLC.




**Scheme 4** Plausible mechanism.


This journal is © the Partner Organisations 2020

Published on 19 March 2020. Downloaded by Fudan University on 2/20/2021 4:53:07 AM.

Based on the previous reports,[16] a possible mechanism and an activation model were proposed in Scheme 4. Initially, the ferrocene-derived bifunctional phosphine attacked **1a** creating the zwitterionic intermediate **Int-I**, which subsequently underwent γ-addition to the more electron-deficient carbon of pyrazoloneyldiene oxindole **2** to give intermediate **II**. Then, **II** was converted into **III** through intramolecular Michael addition. Finally, a proton transfer followed along with the release of the chiral phosphine catalyst and yielded the [3 + 2] annulation product spiro-cyclopentene compound **3** combined with oxindole and pyrazolone.

## Conclusions

In conclusion, we have developed a highly regio-, diastereo- and enantioselective methodology to construct spiro-cyclopentene-linked oxindole and pyrazolone compounds, which bear three consecutive stereocenters and two vicinal spiroquaternary chiral centers (dr > 19 : 1 and ee up to 98%) *via* the enantioselective [3 + 2] annulation reaction of γ-substituted allenoates with pyrazoloneyldiene oxindole by ferrocene-derived bifunctional phosphine **P8** as a catalyst. Efforts to explore a more effective chiral ferrocenylphosphine catalyst and biological evaluations of our synthetic spiro-oxindole cyclopentene-pyrazolones are underway in our laboratory.

## Conflicts of interest

There are no conflicts to declare.

## Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 21978271 and 21706234) and the Natural Science Foundation of Zhejiang Province of China (No. LY19B060011).

## Notes and references

1 For recent reviews see: (*a*) G. S. Singh and Z. Y. Desta, Isatins as privileged molecules in design and synthesis of spiro-fused cyclic frameworks, *Chem. Rev.*, 2012, **112**, 6104; (*b*) K. W. Quasdorf and L. W. Overman, Catalytic enantioselective synthesis of quaternary carbon stereocentres, *Nature*, 2014, **516**, 181; (*c*) E. M. Carreira and T. C. Fessard, Four-membered ring-containing spirocycles: synthetic strategies and opportunities, *Chem. Rev.*, 2014, **114**, 8257. For selected examples see: (*d*) A. V. Velikorodov, V. A. Ionova, O. V. Degtyarev and l. t. sukhenko, synthesis and antimicrobial and antifungal activity of carbamate-functionized spiro compounds, *Pharm. Chem. J.*, 2013, **46**, 715; (*e*) J.-A. Xiao, H.-G. Zhang, S. Liang, J.-W. Ren, H. Yang and X.-Q. Chen, Synthesis of pyrrolo(spiro-[2.3′]-oxindole)-spiro-[4.3″]-oxindole via 1,3-dipolar cycloaddition of azomethine ylides with 3-acetonylideneoxindole, *J. Org. Chem.*, 2013, **78**, 11577; (*f*) Y. Kia, H. Osman, R. S. Kumar, V. Murugaiyah, A. Basiri, S. Perumal, H. A. Wahab and C. S. Bing, Synthesis and discovery of novel piperidone-grafted mono- and bis-spirooxindole hexahydropyrrolizines as potent cholinesterase inhibitors, *Bioorg. Med. Chem.*, 2013, **21**, 1696.

2 For selected examples see: (*a*) Z.-Y. Cao and J. Zhou, Catalytic asymmetric synthesis of polysubstituted spirocyclopropyl oxindoles: organocatalysis versus, transition metal catalysis, *Org. Chem. Front.*, 2015, **2**, 849; (*b*) M. J. James, P. O'Brien, R. J. K. Taylor and W. P. Unsworth, Synthesis of spirocyclic indolenines, *Chem. – Eur. J.*, 2016, **22**, 2856; (*c*) C. R. Reddy, S. K. Prajapti, K. Warudikar, R. Ranjan and B. B. Rao, ipso-Cyclization: an emerging tool for multifunctional spirocyclohexadienones, *Org. Biomol. Chem.*, 2017, **15**, 3130; (*d*) S. Kotha, N. R. Panguluri and R. Ali, First-principles calculations and optical absorption spectrum of a light-colored aluminum substituted ε-iron oxide magnet, *Eur. J. Org. Chem.*, 2017, **2017**, 531; (*e*) G.-J. Mei and F. Shi, Catalytic asymmetric synthesis of spirooxindoles: recent developments, *Chem. Commun.*, 2018, **54**, 6607; (*f*) T.-T. Cong, H.-M. Wang, X.-Z. Li, H.-H. Wu and J.-L. Zhang, Chiral bifunctional bisphosphine enabled enantioselective tandem Michael addition of tryptamine-derived oxindoles to ynones, *Chem. Commun.*, 2019, 9176.

3 For selected examples, see: (*a*) V.-C. Gibson, C. Redshaw and G.-A. Solan, Bis(imino)pyridines: Surprisingly reactive ligands and a gateway to new families of catalysts, *Chem. Rev.*, 2007, **107**, 1745; (*b*) A. Zea, A.-N. R. Alba, A. Mazzanti, A. Moyano and R. Rios, Highly enantioselective cascade synthesis of spiropyrazolones, *Org. Biomol. Chem.*, 2011, **9**, 6519; (*c*) L. Liu, Y. Zhong, P. Zhang, X. Jiang and R. Wang, Core scaffold-inspired stereoselective synthesis of spiropyrazolones via an organocatalytic michael/cyclization sequence, *J. Org. Chem.*, 2012, **77**, 10228; (*d*) J.-X. Zhang, N.-K. Li, Z.-M. Liu, X.-F. Huang, Z.-C. Geng and X.-W. Wang, Enantioselective synthesis of unsymmetrical diaryl-substituted spirocyclo-hexanonepyrazolones through a cascade [4+2]double michael addition, *Adv. Synth. Catal.*, 2013, **355**, 797; (*e*) J. Liang, Q. Chen, L. Liu, X. Jiang and R. Wang, An organocatalytic asymmetric double Michael cascade reaction of unsaturated ketones and unsaturated pyrazolones: highly efficient synthesis of spiropyrazolone derivatives, *Org. Biomol. Chem.*, 2013, **11**, 1441; (*f*) J.-H. Li and D.-M. Du, Organocatalyzed cascade aza-michael/michael addition for the asymmetric construction of highly functionalized spiropyrazolone tetrahydroquinolines, *Chem. – Asian J.*, 2014, **9**, 3278; (*g*) B. Han, W. Huang, W. Ren, G. He, J.-H. Wang and C. Peng, Asymmetric synthesis of cyclohexane-fused drug-like spirocyclic scaffolds containing six contiguous stereogenic centers via organocatalytic cascade reactions, *Adv. Synth. Catal.*, 2015, **357**, 561; (*h*) S. R. Yetra, S. Mondal, S. Mukherjee, R. G. Gonnade and

Published on 19 March 2020. Downloaded by Fudan University on 2/20/2021 4:53:07 AM.

A. T. Biju, Enantioselective synthesis of spirocyclohexadienones by NHC-catalyzed formal [3+3] annulation reaction of enals, *Angew. Chem., Int. Ed.*, 2016, **55**, 268; (*i*) J. Zheng, S.-B. Wang, C. Zheng and S.-L. You, Asymmetric synthesis of spiropyrazolones by Rhodium-catalyzed C(sp2)-H functionalization/annulation reactions, *Angew. Chem., Int. Ed.*, 2017, **56**, 4540.

4 (*a*) T. Masashi, K. Yuu, I. Haruaki, F. Eri, K. Jun, W. Masayuki, A. Kenichi and K. Jun'ichi, Absolute stereochemistry of citrinadins A and B from marine-derived fungus, *J. Org. Chem.*, 2005, **70**, 9430; (*b*) E. Mercado-Marin, P. Garcia-Reynaga and S. Romminger, Total synthesis and isolation of citrinalin and cyclopiamine congeners, *Nature*, 2014, **509**, 318.

5 M. Gicquel, C. Gomez, M. C. C. Alvarez, O. Pamlard, V. G. E. Jacquet, J. Bignon, A. Voituriez and A. Marinetti, Inhibition of p53-murine double minute 2 (MDM2) interactions with 3,3′-spirocyclopentene oxindole derivatives, *J. Med. Chem.*, 2018, **61**, 9386.

6 (*a*) I. Schlemminger, B. Schmidt, D. Flockerzi, H. Tenor, C. Zitt, A. Hatzelmann, D. Marx, C. Braun, R. Kuelzer, A. Heuser, H.-P. kley and G.-J. Sterk, Novel pyrazolone-derivatives and their use as PD4 inhibitors, *Germany Patent*, WO2010055083, 2010; (*b*) B. Schmidt, C. Scheufler, J. Volz, M.-P. Feth, R.-P. Hummel, A. Hatzeimann, C. Zitt, A. Wohlsen, D. Marx, H.-P. Kley, D. Marx, D. Ockert, A. Heuser, J.-A. M. Christiaans, G.-J. Sterk and W.-M. P. B. Menge, pyrazolone derivatives as PDE4 inhibitors, *Germany Patent*, WO2008138939, 2008.

7 I. Tada, M. Motoki, N. Takahashi and T. Miyata, Synthesis and Structure-Activity Relationships of Miticidal 4,5-Dihydropyrazole-5-thiones, *Pestic. Sci.*, 1996, **48**, 165.

8 (*a*) R. Morphy, C. Kay and Z. Rankovic, From magic bullets to designed multiple ligands, *Drug Discovery Today*, 2004, **9**, 641; (*b*) H.-Y. Zhang, D.-P. Yang and G.-Y. Tang, Multipotent antioxidants: from screening to design, *Drug Discovery Today*, 2006, **11**, 749; (*c*) C. D. Duarte, E. J. Barreiro and C. A. M. Fraga, Privileged Structures: A useful concept for the rational design of new lead drug candidates, *Mini-Rev. Med. Chem.*, 2007, **7**, 1108; (*d*) A. Talevi, Multi-target pharmacology: possibilities and limitations of the "skeleton key approach" from a medicinal chemist perspective, *Front. Pharmacol.*, 2015, **6**, 205; (*e*) R. J. Pagliero, M. Kaiser, R. Brun, M. J. Nieto and M. R. Mazzieri, Lead selection of antiparasitic compounds from a focused library of benzenesulfonyl derivatives of heterocycles, *Bioorg. Med. Chem. Lett.*, 2017, **27**, 3945.

9 (*a*) C. Zhang and X. Lu, Phosphine-catalyzed cycloaddition of 2,3-butadienoates or 2-butynoates with electron-deficient olefins. A novel [3+2] annulation approach to cyclopentenes, *J. Org. Chem.*, 1995, **60**, 2906; (*b*) Y. Du, X. Lu and Y. Yu, Highly regioselective construction of spirocycles via phosphine-catalyzed [3+2]-cycloaddition, *J. Org. Chem.*, 2002, **67**, 8901; (*c*) Y. Du and X. Lu, A phosphine-catalyzed [3+2] cycloaddition strategy leading to the first total synthesis of (−)-Hinesol, *J. Org. Chem.*, 2003, **68**, 6463.

10 For recent reviews see: (*a*) L.-W. Ye, J. Zhou and Y. Tang, Phosphine-triggered synthesis of functionalized cyclic compounds, *Chem. Soc. Rev.*, 2008, **37**, 1140; (*b*) B. J. Cowen and S. J. Miller, Enantioselective catalysis and complexity generation from allenoates, *Chem. Soc. Rev.*, 2009, **38**, 3102; (*c*) Q.-Y. Zhao, Z. Lian, Y. Wei and M. Shi, Development of asymmetric phosphine-promoted annulations of allenes with electron-deficient olefins and imines, *Chem. Commun.*, 2012, **48**, 1724; (*d*) Y. C. Fan and O. Kwon, Advances in nucleophilic phosphine catalysis of alkenes, allenes, alkynes, and MBHADs, *Chem. Commun.*, 2013, **49**, 11588; (*e*) Z. Wang, X. Xu and O. Kwon, Phosphine catalysis of allenes with electrophiles, *Chem. Soc. Rev.*, 2014, **43**, 2927; (*f*) W.-B. Li and J.-L. Zhang, Recent developments in the synthesis and utilization of chiral β-aminophosphine derivatives as catalysts or ligands, *Chem. Soc. Rev.*, 2016, **45**, 1657. For selected examples see: (*g*) H. Xiao, Z. Chai, C.-W. Zheng, Y.-Q. Yang, W. Liu, J.-K. Zhang and G. Zhao, Asymmetric [3+2] cycloadditions of allenoates and dual activated olefins catalyzed by simple bifunctional n-acyl aminophosphines, *Angew. Chem., Int. Ed.*, 2010, **49**, 4467; (*h*) X. Han, Y. Wang, F. Zhong and Y. Lu, Enantioselective [3+2] cycloaddition of allenes to acrylates catalyzed by dipeptide-derived phosphines: facile creation of functionalized cyclopentenes containing quaternary stereogenic centers, *J. Am. Chem. Soc.*, 2011, **133**, 1726; (*i*) Q. Zhao, X. Han, Y. Wei, M. Shi and Y. Lu, Enantioselective [3+2] cycloaddition of allenes to acrylates catalyzed by dipeptide-derived phosphines: facile creation of functionalized cyclopentenes containing quaternary stereogenic centers, *Chem. Commun.*, 2012, **48**, 970; (*j*) X.-N. Zhang and M. Shi, Phosphine-catalyzed [3+2] cycloaddition of 4,4-dicyano-2-methylenebut-3-enoates with benzyl buta-2,3-dienoate and penta-3,4-dien-2-one, *ACS Catal.*, 2013, **3**, 507; (*k*) S. Y. Lee, Y. Fujiwara, A. Nishiguchi, M. Kalek and G. C. Fu, Phosphine-catalyzed enantioselective intramolecular [3+2] annulations to generate fused ring systems, *J. Am. Chem. Soc.*, 2015, **137**, 4587; (*l*) W. Zhou, H.-M. Wang, M. Tao, C.-Z. Zhu, T.-Y. Lin and J.-L. Zhang, Phosphine-catalyzed enantioselective [3+2] cycloadditions of γ-substituted allenoates with β-perfluoroalkyl enones, *Chem. Sci.*, 2017, **8**, 4660.

11 (*a*) W.-L. Chan, X. Tang, F. Zhang, G. Quek, G.-J. Mei and Y. Lu, Phosphine-catalyzed (3+2) annulation of isoindigos with allenes: enantioselective formation of two vicinal quaternary stereogenic centers, *Angew. Chem.*, 2019, **58**, 6260; (*b*) J. Zhang, W.-L. Chan, L. Chen, N. Ullah and Y. Lu, Creation of bispiro[pyrazolone-3,3′-oxindoles] via a phosphine-catalyzed enantioselective [3+2] annulation of the Morita-Baylis-Hillman carbonates with pyrazoloneyldiene oxindoles, *Org. Chem. Front.*, 2019, **6**, 2210.

12 For our previous works, see: (*a*) A.-P. Tu, H.-W. Hu, Q. Tang and W.-H. Zhong, Asymmetric [3+2]-cycloaddition of Morita-Baylis-Hillman carbonates with maleimides catalyzed by chiral ferrocenylphosphines, *Synth. Commun.*, 2014, **44**, 3392; (*b*) H.-W. Hu, S.-X. Yu, L.-L. Zhu, L.-X. Zhou

 This journal is © the Partner Organisations 2020

Published on 19 March 2020. Downloaded by Fudan University on 2/20/2021 4:53:07 AM.

and W.-H. Zhong, Chiral bifunctional ferrocenylphosphine catalyzed highly enantioselective [3+2] cycloaddition reaction, *Org. Biomol. Chem.*, 2016, **14**, 752; (*c*) L.-L. Zhu, H.-W. Hu, L. Qi, Y. Zheng and W.-H. Zhong, Enantioselective allylic substitution of Morita-Baylis-Hillman adducts catalyzed by chiral bifunctional ferrocenylphosphines, *Eur. J. Org. Chem.*, 2016, **2016**, 2139; (*d*) W.-J. Luo, H.-W. Hu, S.-F. Nian, L. Qi, F. Ling and W.-H. Zhong, Phosphine-catalyzed [3+2] annulation reaction: highly regio- and diastereoselective synthesis of 2-azaspiro[4.4]nonene-1,3-diones, *Org. Biomol. Chem.*, 2017, **15**, 7523; (*e*) F. Ling, S.-F. Nian, J.-C. Chen, W.-J. Luo, Z. Wang, Y.-P. Lv and W.-H. Zhong, Development of ferrocene-based diamine-phosphine-sulfonamide ligands for iridium-catalyzed asymmetric hydrogenation of ketones, *J. Org. Chem.*, 2018, **83**, 10749; (*f*) W.-J. Luo, D.-G. Song, L. Fang, S.-S. Nian, H.-C. Hou, F. Ling and W.-H. Zhong, Chiral bifunctional ferrocenylphosphine-catalyzed enantioselective allylic alkylation of Morita-Baylis-Hillman carbonates with pyrazolin-5-ones, *Asian J. Org. Chem.*, 2018, **7**, 2417; (*g*) F. Ling, H.-C. Hou, J.-C. Chen, S.-S. Nian, X. Yi, Z. Wang, D.-G. Song and W.-H. Zhong, Highly enantioselective synthesis of chiral benzhydrols via manganese catalyzed asymmetric hydrogenation of unsymmetrical benzophenones using an imidazole-based chiral PNN tridentate ligand, *Org. Lett.*, 2019, **21**, 3937; (*h*) S.-F. Nian, F. Ling, J.-C. Chen, Z. Wang, H.-W. Shen, X. Yi, Y.-F. Yang, Y.-B. She and W.-H. Zhong, Highly enantioselective hydrogenation of non-ortho-substituted 2-pyridyl aryl ketones via iridium-f-diaphos catalysis, *Org. Lett.*, 2019, **21**, 5392.

13 W.-J. Luo, B.-X. Shao, J.-Y. Li, D.-G. Song, X. Yi, F. Ling and W.-H. Zhong, Divergent synthesis of spirocyclopentene-pyrazolones and pyrano[2,3-c]-pyrazoles via Lewis base controlled annulation reactions, *Tetrahedron Lett.*, 2019, **60**, 151206.

14 H.-M. Wang, W.-K. Lu and J.-L. Zhang, Ferrocene derived bifunctional phosphine-catalyzed asymmetric oxa-[4+2] cycloaddition of α-substituted allenones with enones, *Chem. – Eur. J.*, 2017, **23**, 13587.

15 Crystallographic data for the structural analysis of compound **3ag** have been deposited at the Cambridge Crystallographic Data Centre as CCDC 1961000.†

16 (*a*) M. Shi and L.-H. Chen, Chiral phosphine Lewis base catalyzed asymmetric aza-Baylis-Hillman reaction of N-sulfonated imines with methyl vinyl ketone and phenyl acrylate, *Chem. Commun.*, 2003, **39**, 1310–1311; (*b*) C.-W. Cho, J.-R. Kong and M.-J. Krische, Phosphine-catalyzed regiospecific allylic amination and dynamic kinetic resolution of Morita-Baylis-Hillman acetates, *Org. Lett.*, 2004, **6**, 1337; (*c*) H. Park, C.-W. Cho and M.-J. Krische, Phosphine-catalyzed allylic substitution of Morita-Baylis-Hillman acetates: synthesis of n-protected β-aminophosphonic acid esters, *J. Org. Chem.*, 2006, **71**, 7892; (*d*) Y.-Q. Jiang, Y.-L. Shi and M. Shi, Chiral phosphine-catalyzed enantioselective construction of γ-butenolides through substitution of Morita-Baylis-Hillman acetates with 2-trimethylsilyloxy furan, *J. Am. Chem. Soc.*, 2008, **130**, 7202; (*e*) Y. Wei and M. Shi, Multifunctional chiral phosphine organocatalysts in catalytic asymmetric Morita-Baylis-Hillman and related reactions, *Acc. Chem. Res.*, 2010, **43**, 1005; (*f*) H.-P. Deng, Y. Wei and M. Shi, Chiral bifunctional thiourea-phosphane organocatalysts in asymmetric allylic amination of Morita-Baylis-Hillman acetates, *Eur. J. Org. Chem.*, 2011, **10**, 1956; (*g*) Y. Wei and M. Shi, Recent advances in organocatalytic asymmetric Morita-Baylis-Hillman/aza-Morita-Baylis-Hillman reactions, *Chem. Rev.*, 2013, **113**, 6659; (*h*) Y. Xiao, Z. Sun, H. Guo and O. Kwon, Chiral phosphines in nucleophilic organocatalysis, *Beilstein J. Org. Chem.*, 2014, **10**, 2089; (*i*) P.-Z. Xie and Y. Huang, Morita-Baylis-Hillman adduct derivatives (MBHADs): versatile reactivity in Lewis base-promoted annulation, *Org. Biomol. Chem.*, 2015, **13**, 8578; (*j*) T. Wang, X. Han, F. Zhong, W. Yao and Y. Lu, Amino Acid-Derived Bifunctional Phosphines for Enantioselective Transformations, *Acc. Chem. Res.*, 2016, **49**, 1369; (*k*) H.-Z. Ni, W.-L. Chan and Y.-X. Lu, Phosphine-catalyzed asymmetric organic reactions, *Chem. Rev.*, 2018, **118**, 9344; (*l*) H.-C. Guo, Y.-C. Fan, Z.-H. Sun, Y. Wu and O. Kwon, Phosphine organocatalysis, *Chem. Rev.*, 2018, **118**, 10049.